CKD

① Kidneys damaged (DM + HTN most common!)
↓
② healthy nephrons take on more blood flow
↓
③ those nephrons get tired and body ↓ blood flow to kidneys
↓
④ RAAS activation → ⑤ further ↓ blood flow (vasocxn) → ⑥ more damage = more s/s

Normal GFR = 90
stage 5 CKD = GFR < 15
Renal insufficiency = ↑ stage 3/4 = GFR 30-59 (3)
start to have s/s! GFR 15-29 (4)

urea → can't excrete → BUN ⇧ (uremia) → N/V lethargy pruritus uremic fetor

Acid/base → can't lose H^+ (acid) or make HCO_3 → metabolic acidosis → Kussmauls to compensate

Fluid balance → nephrons normally make urine but can't when damaged → oliguria → anuria → hypervolemia, ⇩Na, CHF

Glucose → can't excrete glucose → ⇧BS (glucose) → ↑ insulin → DM → ↑ infection risk

EPO → ↓ EPO → anemia → fatigue, SOB

vitamin D → can't activate → can't absorb Ca → ⇩Ca, ⇧Phos
↳ PTH ↑ → break down bone → osteoporosis, fractures

K/Mg → can't excrete → ⇧K, ⇧Mg

Triglycerides → don't excrete → ⇧triglycerides → CVD

creatinine → can't excrete → ⇧Cr

Proteins → leaky glomerulous → proteinuria → ↓ (blood) oncotic pressure → edema, ↓ BP (early)

RAAS → RAAS constantly activated d/t ↓ blood flow
↓
HTN, ⇧Na, hypervolemia

Normal kidney functions:

- Excrete urea
- Acid/base balance
- Fluid balance
- Glucose regulation (excrete when ↑ BS)
- EPO release
- vitamin D activation
- K + Mg excrete
- Triglyceride excrete
- creatinine excretion
- Normally proteins don't get excreted

- RAAS ↳ ↓ blood flow
↓ renin
↓ angiotensin
↓ angiotensin I
↓ angiotensin II → vaso-cxn ↓
↓ aldosterone ⇧BP
↓
↑ Na/H_2O absorption
K excrete